Introduction

Background

We have noticed that people tend to trash items when they do not have a convenient way to donate or sell them. According to the United States EPA, "Municipal Solid Waste (MSW)—more commonly known as trash or garbage—consists of everyday items… such as product packaging, grass clippings, furniture, clothing, bottles, food scraps, newspapers, appliances, paint, and batteries. This comes from our homes, schools, hospitals, and businesses." There are people that are in need of, or could use, these items.

Problem Statement

We believe there is a need for a better way to reuse items people no longer want, a way to connect these items to the people who need or could use them.

Design Goals

Our main design goals were relatively self-explanatory. They included fast, simple means to post items to donate, fast, simple means to request items you need, efficient method to match a donator and a donatee, provide simple, consistent feedback through the donation process, provide efficient ways to search for items being donated, and increasing availability of resources to people in need.

Usability Goals

Our main usability goals included accessibility, familiarity, safety, efficiency, and engaging. We wanted the application to feel familiar enough to those existing today for social media and selling items to make it easy to use. The app should keep users safe by not having them communicate with each other, share locations publicly, or need to transport their items themselves to a stranger. The app should also be efficient and engaging in the sense that it not only matches those donating with those receiving, but also keeps the user informed, gives good feedback, and makes the user feel like they didn't need to do very much to get the job done, whether that's requesting or donating items.

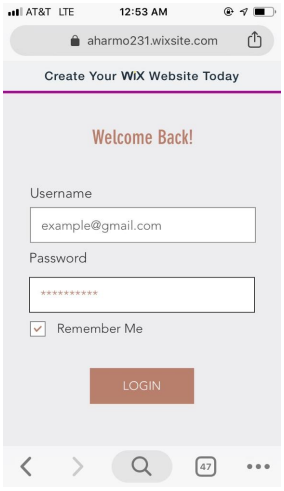
Design description

This section contains a description, rationale, and screenshots of our design choices and app functionality. Any screenshots of pages with forms have only the first section showing, but they can scroll in the application to show the rest of the form sections. Each screenshot is numbered so it can be more easily matched with its description.

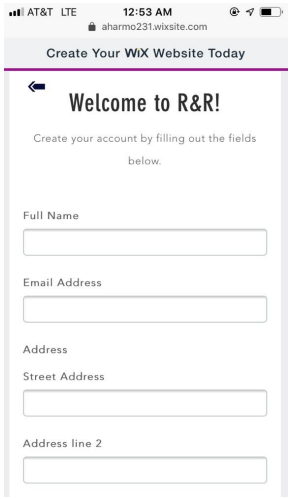
The application has a basic login or sign up page (1) that has become the standard for most mobile and web applications. We show three options: Login, Sign Up, and Learn More. Both Login (2) and Sign Up (3) will take the user to the Home page after completing. Learn More takes the user to the About page (4), giving information on the app itself, why it was created, and generally how it could be used.



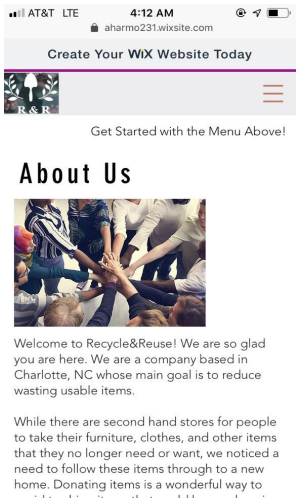
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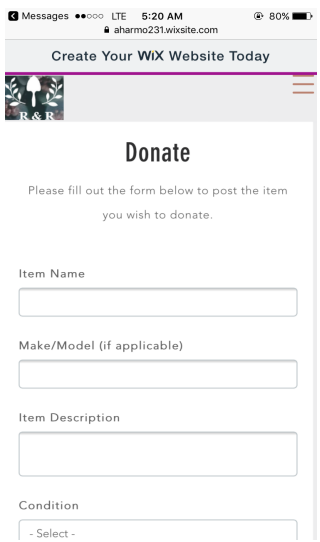


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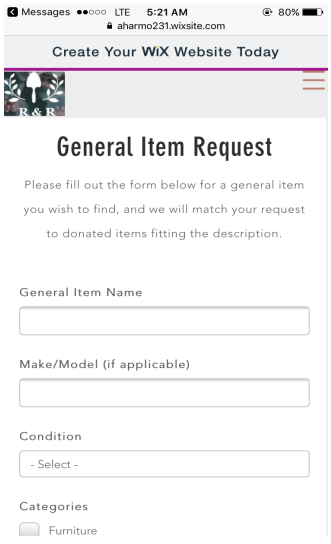
The Home page (5) offers the main functionality of the application: Donate, Request, and Search. Donate takes the user to a form to complete about the item they wish to donate (6). They enter the item name, description, condition, and upload photos. They receive a success message that the item has been posted after submitting. Request takes the user to a short form to request general items (7), for example, a couch, kitchen chairs, or a table. The user could enter general item names or get more specific with make or model information, condition, and categories. This form can be used if the user does not wish to search through items themselves or if the item they want is not listed at the moment. The form would go into the system, and the application would try to find items matching what the user requested. If matches are found, the user is alerted and then has the option to request an item that was a match, or they can choose to ignore it and wait for other matches.



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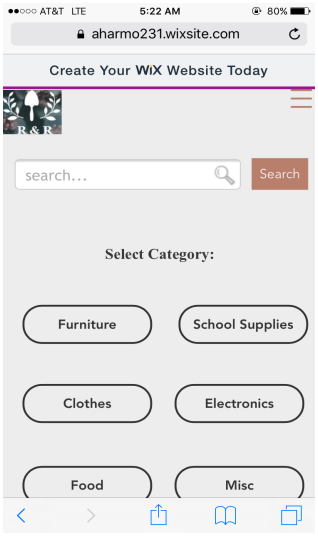


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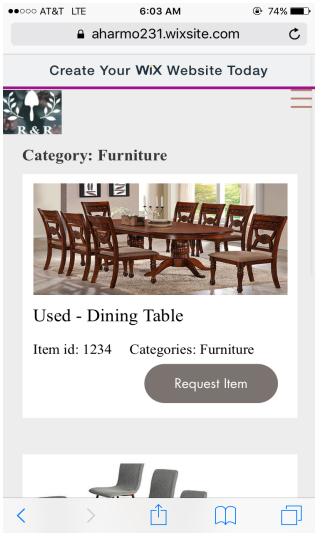


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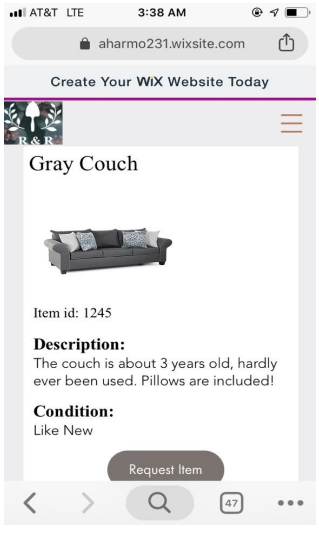
Search takes the user to a page where they can search for items using the search bar or choose a category to search by (8). Using either method will take the user to a list view of items, either results of the item name they typed in or results of items in the chosen category (9). Once there, they can request the item from the list view, or click on the item to view more information about it (10) and then request it from there. Once the user requests an item, they receive a message that the request went through (11).



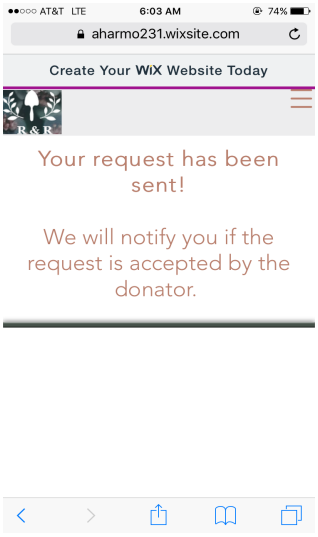
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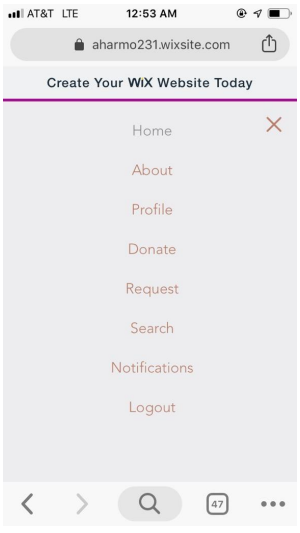


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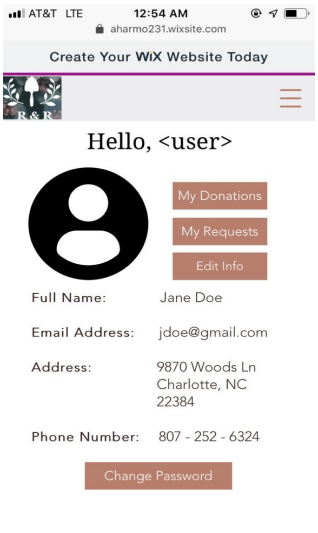


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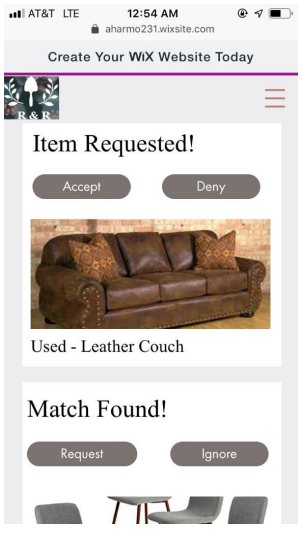
We included a hamburger menu that has the following options: Home, About, Profile, Donate Request, Search, Notifications, and Logout (12). This allows users to not only get to more pages without cluttering up the screen with all the options, but also get to the main functions Donate, Request, and Search without needing to first go back to the Home page. Logout will take the user back to the application's front page with the Login, Sign Up, and Learn More options.



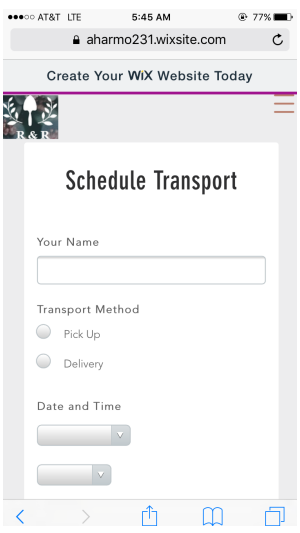
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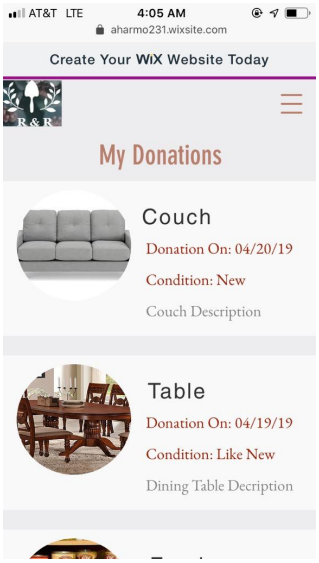
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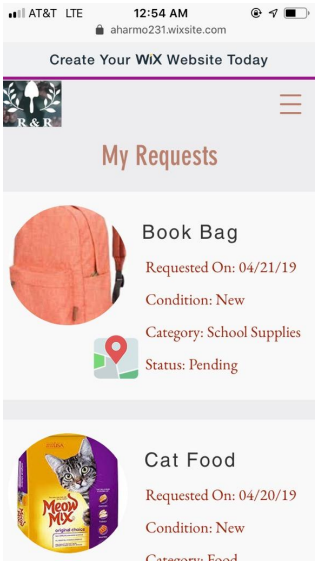
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Profile is where a user can view and edit their information, change their password, as well as navigate to a list of items they have donated and a list of the items they have requested through the My Donations and My Requests buttons (13). Notifications will take the user to a list of things they have been notified about, here for example, someone requesting an item the user donated and also the app finding a match for an item the user requested (14). If someone requests an item that user has donated, the user has the option to accept the request or deny it. If they accept the request, it takes the user to the Schedule Transport page (15). The Schedule Transport page is where the user fills in time, date, and location information for the item to be picked up by a company driver, or if choosing to drop off the item at the warehouse, the user would enter the time, date, and location of when they would drop it off and which warehouse. Once they complete this, the user who requested that item will be alerted that their request was accepted, and then they are taken to the same Schedule Transport page for the item to be dropped off to them by a driver or for them to pick the item up at the warehouse.

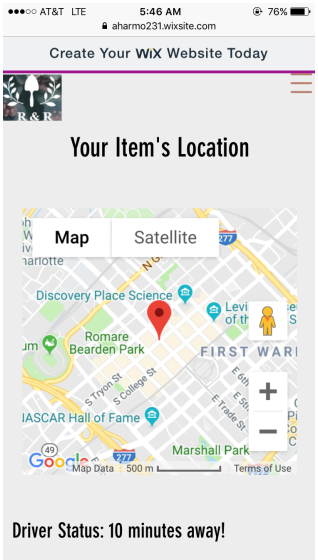
As mentioned previously, My Profile contains buttons to see the user's requests made and items donated. The My Donations page (16) shows the items the user donated, including the item name, donation date, condition, and first line of the description. The My Requests page (17) shows a brief overview of the items that have been requested by that user, including things like the item's name, when the request was made, the condition of the item, the category the item falls in, the status of the item request, and a map logo that can be clicked to track the location of the item, although this is only if the item has been picked up and is on the way or the item has reached the warehouse. It will not give the location of another user who has the item. The Item Location page (18) would be more detailed in the future for an item in transit, showing the name of the driver, car make/model, and a help number to call, in addition to the time and distance the item is away. It would also be accessible from the notifications a user would get when a driver is on their way.



(16)



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(18)

Design Rationale

The reasoning for most of our design choices was to mimic social media apps and apps used for selling second hand items. We aimed to have the ease of scrolling through a social media site, along with the brief forms found on selling apps like forms for uploading items. We kept our Login, Sign Up, and About pages as the standard for the options and information shown. Larger buttons were used throughout the application, as well as having fewer of them to try to keep the look clean and make it simple to use with fewer choices. We kept most of the background coloring to the Login and Home pages to add an earthy tone and give the user the feeling of nature and the purpose of reusing items. This background was not used throughout the application because it would clash with the forms and scrolling features too much. We did not want it to look busy or cluttered. So, we stuck to a white background and just left the logo at the top left hand corner having a piece of the background from the login and home pages behind it as part of the logo.

The bar at the top of the screen containing the logo and hamburger menu was left gray to be different from the white background of the sections that were scrollable, but not to stand out too much with bright color. The hamburger menu was colored orange, brown to match the main buttons' color throughout the app. We chose the orange, brown for the buttons because it was a color that draws attention to those button options while still keeping it somewhat earthy with the brown in it. We chose the hamburger menu because we felt this was the cleanest way to offer all of the page options listed and not have things be in the way. We left the notification section in the hamburger menu rather than having its own icon at the top of the page to avoid clutter. We planned to have the hamburger menu show a small number next to it and have the lines bolded if there were notifications. This way, it would still draw the user's attention, but it wouldn't need to be its own icon taking up space at the top.

Photos of items in the list view were left large enough to see items pretty well without clicking on them, but small enough to leave room for basic descriptive information to be shown next to the image. We wanted users to be able to get most of the information they could need or be looking for without having an extra click if they don't want to see the item page itself. We left out any messaging tools or icons, with the main difference of our app from those existing being that users do not need to communicate with strangers to donate or find items.

Discount Evaluation

We received evaluations from two groups in class of our low-fidelity paper prototype for our mobile app. This was the one shown in part two of the project, as the last design concept in the presentation. Group one felt that some buttons might be slightly small for users, especially older users or those with larger fingers. The scroll bar on the side that was presented also seemed a bit small, which would present difficulties clicking it and holding down to scroll. They felt that it was missing confirmation messages for certain activities, such as submitting forms and requesting items. They suggested giving

an overview of the application and how it works on the home page. It was also mentioned that it may be fun to include stars for rating the condition rather than words.

The second group suggested adding an option for the user to filter by distance rather than only having it as a built in function to have a specific radius when a user sets their location. They also thought there should be a clarification of the picking up and dropping off methods to pick from, as well as maybe changing the layout and location of pending requests. They said the application was clean and simple in its overall layout though, and they appreciated the button sizes and placements. Our click-through prototype took care of many of the concerns found with the paper prototype. A main example of this is that a user can scroll anywhere on the screen rather than having a scroll bar, so we did not need to worry about the size of the scroll bar that was shown in the paper prototype.

Evaluation Methodology

Our Evaluation Goals

The goals of the evaluation involved seeing how well the app meets our usability and design goals. We wanted the app to overall be efficient, accessible, engaging, safe, and familiar. We wanted the app features to be fast and simple to use. The results from post interview questions we asked were the main way to gauge whether or not we met our usability and design goals.

Planned Measurements and Metrics

The measurements and metrics we used during the tasks given to participants were the number of clicks/presses it took to perform tasks, the time to complete overall tasks, the time each task takes, and the thoughts spoken out loud when performing the tasks. The number of clicks/presses it takes to perform each task is used to gauge how well we meet our usability goal, efficiency, and how well we meet our design goals to have features be simple and quick to use.We used the clicks/presses to gauge how efficient the app was when performing certain tasks and used it to gauge how simple it was to perform certain tasks. The time it took to complete each task was also used to gauge efficiency and it was used to gauge how fast and simple it is to use the main features of the app. The time and clicks were also something we could use to compare with the age of the participant and their experience on apps in general to see if this had any effect. The thoughts spoken out loud during tests were used to gauge how well we met any of the usability and design goals. Since the thoughts spoken out loud could pertain to anything, they were not limited to specific goals when they were evaluated, they just helped us know when a participant needed to work out something that may have been confusing or just their thoughts while performing the tasks. We did not ask participants to think out loud, but we recorded video of each participant using the app, so that if they did say things throughout, we could record it and work it into our evaluation.

The Tasks and Instructions We Gave

We used a video recording of the participant's hand interacting with their phone screen when performing our usability tests. We would have the participants perform specific tasks and record clicks, time, issues, or thoughts expressed. Afterwards, we wanted them to comment on anything they felt should be commented on whether it be something that was frustrating them or something that they liked. The specific tasks we gave them involved them using two of the main features of the app, and then exploring the entire app. We gave them three tasks to perform, two with scenarios prefacing the task, one without just as a general new user looking around:

1. Scenario: You are going to move and need to downsize your furniture. You decide you want to get rid of your couch. You decide you don't care about making a profit, you just want to give it to someone to avoid the trash, but you are unable to transport the couch yourself anywhere. You heard about Recycle&Reuse and decided to try donating your living room couch with it.

   Task: Login and donate a couch with the following information.

   username = bob@gmail.com
   password = pass

   The couch is a small, brown leather indoor couch in good condition. It has rarely been used and is about 3 years old.
   For the purposes of this test, don't worry about uploading a picture of the couch for the donation posting.

2. Scenario: You just moved into your new house and you would like a couch for your living room. You spent a lot of money on the house and moving expenses, so you don't have enough money to buy a couch. You decide to go on the Recycle&Reuse app to find a couch that someone has donated.

   Task: Login to search for and request a couch, using the following information.

   username = bob@gmail.com
   password = pass

   The couch seats 3 people, is dark gray, and in good condition.

3. Task: Look around to see how many of these areas you can find in 3 minutes.
   Place a check mark next to those found:

___ About page	___ Track location of requested item
___ Logout	___ Change password
___ My Donations	___ My Requests
___ Edit Profile	___ Notifications
___ General Item Request	___ Schedule Transport

We felt that having them use the donate and request feature would allow us to gauge how well we are meeting our design goal of having those features of the app be fast and simple to use. Having them try to find all of the pages on the app would allow us to gauge how efficiently the user could navigate the app and how familiar the app is to the users.

Interview or Survey Questions

We had participants do a survey before they performed the tests and then we interviewed them after they were done with the tests. We gave the survey before the test to get an idea of what the participant's age is, as well as their experience with apps, selling or donating items, looking for second-hand items, etc. We wanted to figure out how they donated and received items in the past and any issues they may have had with it. We gave interview questions after the test to get the participants overall experience with the app. The results from the interview questions are mainly used to gauge how well we met are design and usability goals. Here is a screenshot of the survey we gave to participants before the test:

Pre-Study Questions

1) **How many hours a day would you say you're using apps on your phone?**
   ____ Less than 1 hr ____ 1-3 hrs ____ 3-5 hrs ____ 5-7 hrs _____ > 7hrs

2) **What kind of apps do you use?** *(check all that apply)*
   ___ Social Media apps
   ___ Gaming apps
   ___ Messaging apps
   ___ Other, please specify: _____

3) **How much experience do you have selling items on apps?**
   ____ none ___ very little ___ some ___ a lot of experience

4) **How much experience do you have donating items?**
   ____ none ___ very little ___ some ___ a lot of experience

5) **What method did you use to donate?** ***(if applicable)***
   ____ Drop off at second hand store or donation center
   ____ Item picked up by donation center or second hand store
   ____ Gave to a friend or family member
   ____ Other, please specify: _____

6) **Are there any issues you've had donating items?**
   (ex. time, transportation, etc.)

7) **How often would you say you look for second hand items?**
   _____ never ___ occasionally ___ often ___ all the time

8) **Are there any issues you've had finding and/or buying second-hand items?**
   (ex. time, transportation, money, etc.)

9) **Your age:**
   ___ 18-22 ___ 22-29 ___ 30-45 ___ 46-59 ___ 60+

Here are the six interview questions we asked participants after the tests verbally, recording the audio for their responses:

- How would you describe your overall experience with the app?
- Opinions of the design, ease of use, purpose?
- Were there things you expected that weren't there - was this good or bad, why?
- Were there things you didn't expect that were there - was this good or bad, why?
- Were buttons, links, and icons self-explanatory? Could they improve in any way?
- How does our app compare to other apps you've used?

Recruitment

Participants were recruited in a couple ways. One way was through friends and family members to get the older demographic. We simply reached out via text, email, or phone call to them and they agreed to be a participant. The second way was attending Carolina Con. We received permission to sit outside the conference room doors where talks were being held, and asked people passing by if they had a few minutes to test out a prototype. We were able to get the most participants that way. We had 18 participants in total.

Evaluation Results

Pre-Study Survey Results

How many hours a day would you say you're using apps on your phone?

- 10 out of 18 participants spend about 1-3 hours a day using apps

What kind of apps do you use? (check all that apply)

- 16 out of 18 people said messaging apps
- 15 out of 18 people said social media apps

How much experience do you have selling items on apps?

- 12 out of 18 people had no experience using with selling items on apps

How much experience do you have donating items?

- 8 out of 18 people said they have some experience donating other
- Results were varied for this question

What method did you use to donate? (if applicable)

- 15 out of 18 participants drop off at second hand store or donation center
- 11 out of 18 donated to a friend or family member

Are there any issues you've had donating items? (ex. time, transportation, etc.)

- 3 out of the 10 participants that answered this question said time
- 5 out of the 10 participants that answered this question said transportation

How often would you say you look for second hand items?

- 11 out of 18 participants said occasionally

Are there any issues you've had finding and/or buying second-hand items?

- 2 out of the 8 participants that answered this question said time
- 2 out of the 8 participants that answered this question said transportation

Your age:

- 9 out of 18 participants were between the ages of 23 and 29
- Other ages were varied

Usability Test Results

Task 1: Login and donate a a small, brown leather indoor couch in good condition

- The average amount of clicks it took to perform this task was 4.06 or 4 clicks

Task 2: Login to search for and request a couch that seats 3 people, is dark gray, and is in good condition

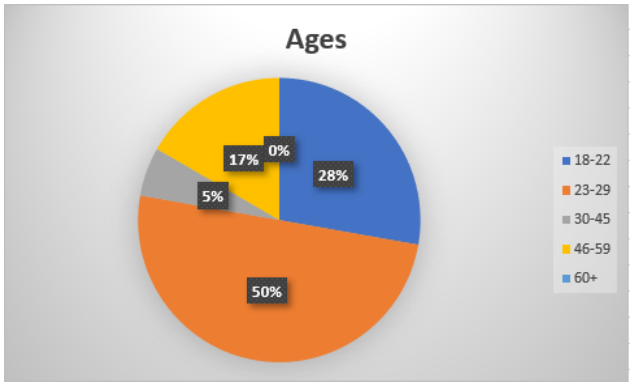
- The average amount of clicks it took to perform this task was 5.06 or 5 clicks

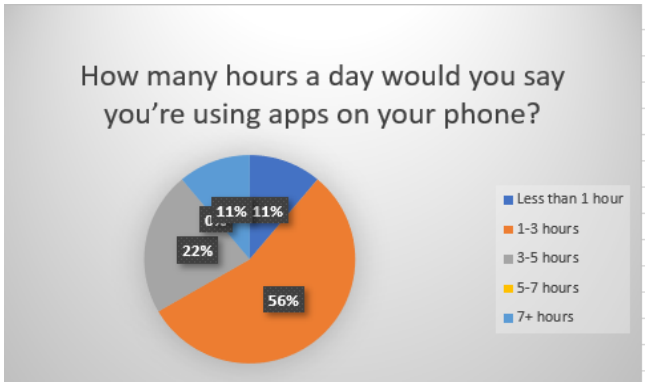
Task 3: Look around to see how many pages/areas you can find in 3 minutes

- 18 out of 18 participants were able to find: About, Logout, My Donations, and Edit Profile
- 5 out of 18 participants were able to find: Schedule Transport

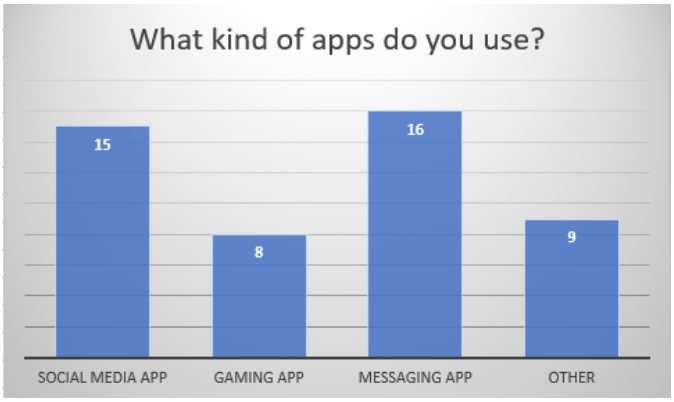
Pre-Study Survey Results

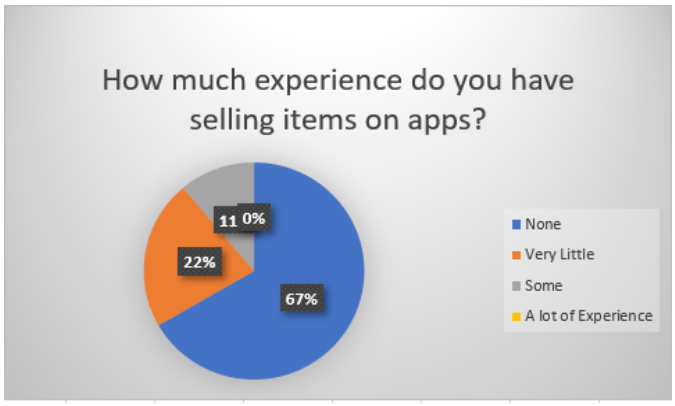
A majority of participants were between the ages of 23 and 29, and they spend one to three hours a day using apps.



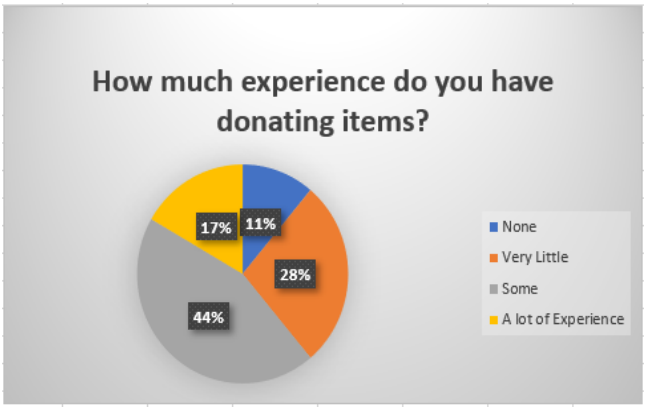


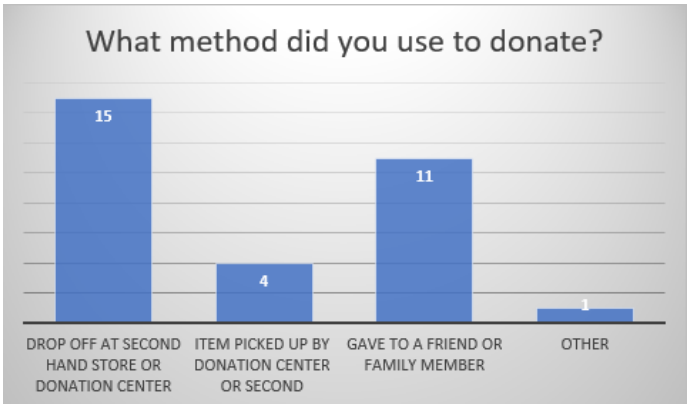
Most people used messaging apps and social media apps, but had no experience selling items using apps. Results varied with how many people have experience donating items. Most people said they have some experience donating items.

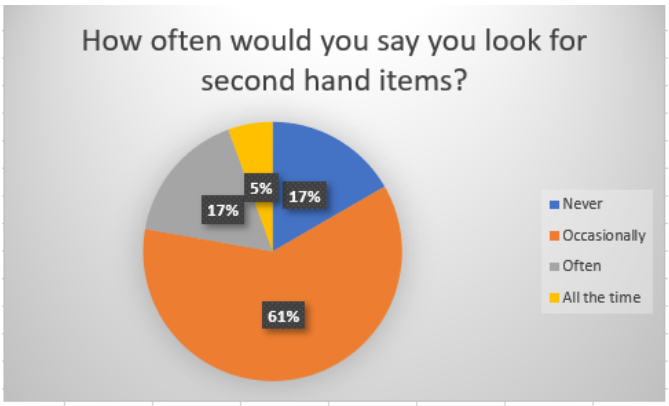




A majority of people said they occasionally look for second and items. A majority of people also said that they either drop off their items at a second hand store or donate their items to a friend or family members, but that the main issues they had with donating or finding and buying second-hand items was time or transportation.







Here is a word cloud of some of the common descriptions found from the audio recordings for the previous questions.



Tasks Results Summary

It takes about 4 clicks for participants to donate an item, including logging in. This seemed to be the simplest and most straight-forward feature. It takes about 5 clicks for participants to search for and request an item, including logging in. Clicks varied if users clicked the Request button, taking them to the General Request Form, first and had to go back and click Search instead. Most participants used the category to search by instead of the search bar, making it much faster for them to find the couch. It was easy for participants to find most of the pages in the app. It was the easiest participants to find About, Logout, My Donations, and Edit Profile. It is harder for participants to find Schedule Transport and Track Location, things that were more clicks away.

Post-Study Interview Question Results Summary

Most people described the app as easy to use, self-explanatory, straightforward, and intuitive. Many said it was similar or good compared to other apps they were familiar with, a couple people saying it was easier to use than some of the apps they had experiences with. Most people said the overall design is good and visually appealing. Multiple people appreciated the track location feature but had trouble knowing they could click the map icon to get there. Multiple people thought there should be some sort of help page that explains how to use the app and its features and had issues with where my requests, my donations, and schedule transport are located. They said the pages were harder to find and didn't expect them to be in the places they were for those pages. A few people said that there should be an option to search by region and to make it clear that the searches are close to their area by default. Most people said no to questions 3 and 4 that were asking if they expected or didn't expect a feature, saying it was what they would have expected, given what they know the app is supposed to do.

How the Results Relate to Our Usability and Design Goals

The results that related most to our usability and design goals were the results we got from the post interview questions. The quantitative data gathered from the tasks are more related to our design goals due to them showing speed and efficiency when using specific features. We break this down in the next section.

Evaluation Implications

Usability Goals

It is hard to say how well we met accessibility since it was not used by more than a small sample of people, but the prototype was easily accessed with the site link we sent to each participant to view on their mobile device. We met our goal of familiarity very well, as almost all participants said it felt like other apps they had used which made it easier to figure out when using it for the first time. Many pointed out the hamburger menu was one that they were used to using for the go-to location to find everything, so it made sense to them that it was the same with our app. Theoretically, we met our goal of safety very well, simulating in the usability studies that a user does not need to go anywhere to

deliver or pick up an item, unless they want to from the warehouse. We also met this goal in the fact that user profiles weren't shown publicly anywhere, and users did not interact with each other. These two things made participants happy, and showed an advantage over other apps used to list second-hand items currently in existence.

We did not meet our goal of efficiency as well as we intended, but it was met for the most part. Most of the feedback said our main functionality was very straightforward and it felt like they didn't have to do much work to get their tasks accomplished. However, the location of some items seemed to make things less efficient when trying to find certain pages or complete tasks like tracking the location of a requested item. We did meet our goal of engaging pretty well, according to the feedback. The success messages shown after completing a task like donating or requesting were encouraging to the users, and scrolling and finding buttons was quick and easy, encouraging exploration. Participants seemed to spend more time exploring even when the study was over out of curiosity, and the fact that they could simply go back to the previous page if they made an erroneous click somewhere, made them feel in control and have the patience to keep trying to find things they were looking for. That also helped meet our efficiency goal a bit more.

Design Goals

Fast, simple means to post items to donate, was met fully. The time it took each participant to donate an item was relatively the same, even for older users that had trouble with other tasks and finding things in general. The donation form was one click away once the participant was signed in, and it was labeled appropriately, so they had no trouble finding it. The form itself was intuitive and the participants moved through it quickly. Fast, simple means to request items you need was not met as well. This had to do with the initial confusion of the difference between the search and request buttons. Most participants clicked on the request button when asked to search and request a specific couch. That took them to the general request form, which wasn't necessarily the wrong way to do things, but it wasn't what we asked for. It took time to explain this and have them go back and use the search option. Then requesting an item was still a couple clicks away. So, we wouldn't say our app ended up satisfying fast and simple means for requesting fully yet.

Efficient method to match a donator and a donatee was hard to determine if it was met. Since the actual functionality does not exist yet, the matching process cannot be fully judged for its efficiency. However, in our prototype, we simulated getting notifications for a match and a request, and this seemed to be efficient in the eyes of our participants. Provide simple, consistent feedback through the donation process was met pretty well in our prototype. There was a success message at the end of the donating process, and participants seemed satisfied with this feedback. No one complained of needing any more feedback for this task.

Provide efficient ways to search for items being donated was met for the most part after participants knew the difference between request and search. Most participants used the categories to click to search a general category of items and found it quick and easy. It could be improved with clickable ways to narrow the search once in the list view. Increasing availability of resources to people in need is hard to say if it was fully met or not with just a prototype, but with our simulation, it seemed encouraging to the participants who noted problems trying to find second-hand items they needed. So, we think it did meet this goal for the most part.

Modifications

We would modify the design in a few ways to better meet our goals. When it comes to efficiency, we need to make some changes to ensure the user knows how the app works before trying to use it, or as they run into issues. Initially, when a user creates an account and is logged in for the first time, we could have a quick tutorial that walks the user through the different features and pages. In addition to this, we could add a How-To page that would appear in the menu, and we would have a help button that is consistent across all pages. We could also add an information button on pages like Donate, Request, and Search. This way, when a user is trying to search for items but clicks on the request option, they can click on the information button to learn what the General Request Form is used for. This would also help for meeting the engaging usability goal.

With the feedback received from participants, we would make My Requests and My Donations more accessible and easy to find, possibly adding My Requests as an option on the Request and Search pages and adding My Donations to the Donate page. This way, if the user is looking for items and decides they want to see what they have already requested, they are one click away. This would be the same for donating the items. We would also improve upon the searching capabilities. The categories made searching for certain kinds of items relatively easy, but some of our participants mentioned how Amazon allows users to click check boxes that narrow down their search results depending on what they are currently searching for. We could add these to the list view of search results.

It was also not clear to many of the participants that the items are automatically sorted by distance to the user based on the address they enter with their account. While many participants felt this was great, some expressed a desire to go outside of their region if they don't find what they are looking for. We could add a list of distance options and have the set range around the user automatically checked but allow other options to increase the distance that the user could click. This would need a pop up about the transportation service possibly not being offered outside of the original range, and the user could click "I understand" or "Cancel."

We can also improve upon the overall design with adding some more color without making things look busy. Most participants liked the colors, but the gray bar at the top could definitely be improved. The hamburger menu could also be a bit larger for our older users, as well as increasing the font size of the

menu options when the menu is expanded. The options are a bit small compared to the other button sizes used across the app. Icons were self-explanatory and our participants felt they were all familiar with them and knew how to use them. The only one in question with many participants was the location tracking icon. It was a picture of a map with a location marker on it, and it wasn't clear to many of the participants that it could be clicked. We could change this to a button with the words "Track Location" instead, as many apps have. This would make users more familiar with it and be able to access this functionality faster.

Lastly, there was a comment that the prototype felt like viewing the mobile version of a website, not a mobile app itself. This made sense with us using Wix to create our prototype. Things like the login or home page scrolling slightly did not feel like an app. We could adjust the screen length and keep things in place more on pages that are not forms or lists. It was also mentioned that the gestures allowed were only scrolling and clicking, which also made it feel more like a website. We could allow swiping gestures to go back or forth, similar to how Facebook does.

In general, we can edit item list layouts as well. Participants did seem happy with everything else they saw. Many noted that it reminded them of apps like LetGo, OfferUp, Craigslist, etc., but that not needing to communicate with users was a great benefit. Overall, we were told the design and options listed made sense for what the app was meant to do. We would just start by making adjustments to those things mentioned previously and see how that changes people's experience.